Cross-Validation in Practice (Summary)

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Standard Validation (Single Split)

- Split data once into three parts:
  - **Training set** — used to fit models.
  - **Validation set** — used to tune hyperparameters and choose the best model.
  - **Test set** — held out until the very end to estimate generalization error.
- **Pros:**
  - Simple to implement and fast.
  - Mimics the “future data” scenario with a true test set.
- **Cons:**
  - High variability — results depend on a single random split.
  - Wastes data, since the test set never contributes to training.
- **Use case:** Large datasets where a single split still leaves plenty of training data.

1. Standard Cross-Validation

- Split data into K folds.
- Rotate: each fold is validation once, others are training.
- **Uses:**
  - Compare multiple candidate models (choose best).
  - Estimate error of a single fixed model.
- **Key idea:** Every observation is used for both training and validation across iterations.

2. CV + Test Dataset

- Hold out a test set from the start.
- Run CV only on training set to choose/tune model.
- After choosing, refit the final model on the entire training set
- Predict and evaluate the error of the test set.
- **Pros:**
  - Honest estimate of generalization error.
  - Avoids reusing the same data for both tuning and final error.
- **Cons:**
  - Test set must be large enough to be reliable.
  - Less data available for training.

3. Nested Cross-Validation

- Outer loop: split into K_{outer} folds.
- For each outer training fold: run an **inner CV** to choose/tune the model.
- Evaluate tuned model on the outer validation fold.
- Average outer validation errors for the overall estimate.
- **Pros:**
  - Uses all data for both tuning and error estimation.
  - Honest error estimate (no leakage).
- **Cons:**
  - Computationally expensive.
  - Harder to explain and implement.

4. Stratified Cross-Validation

- Ensure folds preserve outcome distribution.
  - For classification: same class proportions per fold.
  - For continuous outcomes: bin into quantiles, then stratify.
- **Use:** Reduce variability of error estimates, especially with imbalanced outcomes.
- **Note:** This approach can be used to estimate the error of a single model or combined with *CV + Test Dataset* or *Nested CV*.

5. Group-Specific Error Estimation

- After CV, compute error separately within subgroups.
  - e.g., Age bands, sex, clinical strata.
- **Use:** Check fairness, detect whether the model works equally well across groups.
- **Note:** This approach can be used to estimate the error of a single model or combined with *CV + Test Dataset* or *Nested CV*.

Key Takeaways

- Always separate the roles of training, validation, and testing.
- Standard CV is fine for model selection or error estimation.
- If you want an honest estimate of prediction error *after model selection*, use either:
  - CV + independent test set, or
  - Nested CV.
- Use stratification to stabilize results, and report group-specific errors to evaluate fairness.